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Embedded AI

Project

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Task Description

Preface

Summary

Abstract (English)

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Abbreviations

EMC

Electromagnetic Compatibility

1 Introduction

1.1 Background of the Assignment

1.2 Motivation

1.3 Limitations

1.4 Contributions

1.5 Thesis Structure

2 Literature Study

What has been done by others to solve our or similar problems?

2.1 Finding 1

2.2 Finding 2

2.3 Summary of the Literature Study

3 Theory (Optional)

3.1 Subtopic Example

4 Proposed Solution / Design

4.1 Functional Specification

4.2 Architecture / Design Details

5 Implementation

5.1 Key Implementation Details

6 Testing and Results

6.1 Experimental Setup

6.2 Results

Metrics description:

- **Intersection over Union (IoU):** IoU plays role of evaluating an object localization. It quantifies the overlap between a prediction and ground truth bounding boxes by calculating the ratio of the intersection (overlapping area) to the union (total area covered by both boxes). IoU is a building block for calculating mAP. The resulting score is a value between 0 and 1, which quantifies how accurately a model has located an object in an image. A score of 1 represents a perfect match, while a score of 0 indicates no overlap at all.
The key part of using IoU is to define "*IoU threshold*". This threshold is a predefined value (e.g., 0.5) that determines whether a prediction is correct. If the IoU score for a predicted box is above this threshold, it is classified as a "*true positive*". If the score is below, it's a "*false positive*". This threshold directly influences other performance metrics like Precision and Recall.
- **Precision and Recall:** Precision defines the proportion of true positives among all positive predictions, measuring the ability to avoid misclassification. On the other hand, Recall calculates the proportion of true positives among all actual positives, measuring the model's ability to detect all instances of a class.
- **Average Precision (AP):** AP computes the area under the precision-recall curve, providing a single value that encapsulates the model's precision and recall performance.
- **Mean Average Precision (mAP):** mAP extends the concept of AP by calculating the average AP values across multiple object classes. This is useful in multi-class object detection scenarios to provide a comprehensive evaluation of the model's performance.
- **F1 Score:** The F1 Score is the harmonic mean of precision and recall, providing a balanced assessment of a model's performance while considering both false positives and false negatives.

Table 6.1: Validation summaries.

Table 6.2: YOLO8n

Class	Images	Instances	Box(P)	R	mAP ₅₀	mAP ₅₀₋₉₅
all	98	363	0.927	0.951	0.964	0.733
machine_screw	25	123	0.931	0.967	0.989	0.625
miscellaneous	14	16	0.754	0.812	0.845	0.562
nut	16	41	0.988	1.000	0.995	0.777
washer	20	43	1.000	0.991	0.995	0.906
wood_screw	37	140	0.960	0.986	0.994	0.794

Speed per image (GPU): preprocess 4.5 ms; inference 6.0 ms; loss 0.0 ms; postprocess 2.5 ms.

Model summary (fused): 72 layers, 3,006,623 parameters, 0 gradients, 8.1 GFLOPs

Table 6.3: YOLO11s

Class	Images	Instances	Box(P)	R	mAP ₅₀	mAP ₅₀₋₉₅
all	98	363	0.939	0.934	0.966	0.746
machine_screw	25	123	0.877	0.983	0.978	0.649
miscellaneous	14	16	0.899	0.688	0.867	0.565
nut	16	41	0.970	1.000	0.995	0.805
washer	20	43	0.973	1.000	0.995	0.909
wood_screw	37	140	0.978	1.000	0.995	0.802

Speed per image (GPU): preprocess 4.5 ms; inference 9.8 ms; loss 0.0 ms; postprocess 3.7 ms.

YOLO11s summary (fused): 100 layers, 9,414,735 parameters, 0 gradients, 21.3 GFLOPs

Table 6.4: YOLO11n

Class	Images	Instances	Box(P)	R	mAP ₅₀	mAP ₅₀₋₉₅
all	98	363	0.921	0.935	0.964	0.739
machine_screw	25	123	0.844	0.935	0.957	0.611
miscellaneous	14	16	0.922	0.741	0.888	0.608
nut	16	41	0.970	1.000	0.995	0.799
washer	20	43	0.996	1.000	0.995	0.902
wood_screw	37	140	0.873	1.000	0.983	0.773

Speed per image (GPU): preprocess 0.5 ms; inference 5.6 ms; loss 0.0 ms; postprocess 0.9 ms.

100 layers, 2,583,127 parameters, 0 gradients, 6.3 GFLOPs.

7 Discussion

7.1 Discussion Points

8 Conclusion

9 Future Work

- Item one
- Item two
- Item three

Bibliography

- [1] Redmon, Joseph, et al. "You only look once: Unified, real-time object detection." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.
- [2] Lin, Tsung-Yi, et al. "Microsoft coco: Common objects in context." *European conference on computer vision*. Cham: Springer International Publishing. 2014.

A Text (Appendix A)

B Text (Appendix B)